

DATS6312 – Project Proposal

Group 2

1. Problem Selection

We have chosen the problem of Medically-Aware Recipe Generation using natural language processing and retrieval-augmented generation (RAG). This problem provides an opportunity to combine deep learning and retrieval-based techniques to solve a real-world need—generating personalized meal plans and recipes tailored to individuals with medical conditions, allergies, and dietary goals.

Traditional recipe generators lack medical context, which can be critical for individuals with chronic conditions such as diabetes, hypertension, PCOS, or food allergies. Our solution will intelligently generate recipes that consider both health constraints and user preferences, using a language model guided by medical and nutritional knowledge.

2. Dataset

We will use two main sources of data:

- Medical Knowledge Base: We will collect and scrape data from reputable sources such as MedlinePlus, CDC, EatRight.org, and USDA FoodData Central to build a corpus of dietary and medical guidelines.
- Recipe Dataset: We will use the Kaggle Recipe Dataset (2M+ recipes). We will preprocess and tag this dataset based on attributes like high-protein, low-carb, diabetic-friendly, and allergen content (e.g., nuts, dairy, gluten).

Both datasets will be embedded and indexed using FAISS and Pinecone to support dual-source retrieval in the RAG pipeline.

3. Deep Network

We will use the Meta LLaMA 3.2 (8B or smaller version) as our language model. LLaMA will be used in a RAG setup, where medical and recipe-specific contexts are retrieved and merged to form a prompt that guides the model in generating context-aware and personalized recipes.

If needed, we will use LoRA (Low-Rank Adaptation) to fine-tune the LLaMA model on prompt templates and recipe generation tasks.

4. Framework

Our application will be built using the following technologies:

- PyTorch and Transformers (for model loading and inference)

- SentenceTransformers and FAISS/Pinecone (for building the retrieval system)
- Streamlit (for creating an interactive web app interface)

5. Reference Materials

To build a robust system, we will refer to:

- Meta LLaMA and Hugging Face documentation for model handling
- Pinecone and FAISS documentation for vector retrieval
- Research papers and documentation on Retrieval-Augmented Generation (RAG)

6. Performance Metrics

To evaluate the effectiveness of our system, we will use:

- Relevance of Generated Recipes: Assessed manually based on alignment with medical guidelines and user constraints

7. Schedule

Week 1:

- Literature review and dataset exploration
- Scraping medical data and tagging recipe data

Week 2:

- Building FAISS/Pinecone indices
- Developing RAG pipeline and integrating LLaMA

Week 3:

- Building and testing the Streamlit web app
- Finalizing the report and preparing the project presentation